

BEVERLY, MA, USA

WMO: 725088

Lat: 42.584N

Lon: 70.918W

Elev: 108

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54733

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.9	9.4	-12.6	2.8	8.1	-7.8	3.6	12.2	26.1	24.5	24.1	25.4	10.3	310	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	89.7	73.2	86.2	72.1	83.0	70.4	76.0	85.4	74.4	82.5	72.8	80.1	10.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	122.8	80.6	72.0	118.6	79.2	70.4	112.2	77.4	39.4	85.4	37.9	82.1	36.5	80.1	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	19.3	17.3	-1.5	94.4	6.5	3.1	-6.2	96.6	-10.0	98.4	-13.6	100.2	-18.3	102.5
			-2.7	78.6	5.6	1.2	-6.7	79.5	-9.9	80.3	-13.1	81.0	-17.1	81.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	27.4	29.6	36.2	46.9	56.3	65.5	71.7	70.5	63.5	52.3	42.7	33.5
	DBStd	17.31	10.66	9.21	8.99	8.15	7.66	7.28	5.77	5.74	7.08	7.78	8.42	9.11
	HDD50	2716	701	574	440	152	19	0	0	0	1	65	248	516
	HDD65	6149	1165	992	895	545	290	81	10	15	110	401	669	976
	CDD50	2640	2	2	11	61	216	464	672	636	407	135	29	5
	CDD65	596	0	0	0	4	22	95	217	186	66	6	0	0
	CDH74	4777	0	0	4	64	237	785	1774	1409	463	41	0	0
	CDH80	1446	0	0	0	18	71	236	580	415	124	2	0	0
Wind	WSAvg	7.9	9.0	9.0	9.5	9.0	7.8	7.1	6.6	6.3	6.5	7.5	8.0	8.5
Precipitation	PrecAvg	48.1	4.0	3.5	4.5	4.1	3.7	3.9	3.4	3.6	4.0	4.5	4.1	4.6
	PrecMax	63.7	6.3	8.4	16.8	10.6	16.6	10.8	8.6	8.6	8.3	12.6	8.0	7.5
	PrecMin	33.7	1.3	1.7	0.7	0.5	0.7	0.2	1.4	1.2	1.7	0.2	1.1	1.5
	PrecStd	7.2	1.2	1.5	3.3	2.2	3.0	2.5	1.9	1.9	2.0	3.1	1.6	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.6	60.6	70.0	82.1	87.5	91.2	93.4	92.7	90.0	79.2	69.6	63.0
		MCWB	56.7	53.4	54.6	62.1	66.8	74.4	75.9	75.7	73.3	65.8	61.2	57.2
	2%	DB	52.6	52.4	60.8	72.2	80.8	86.2	89.4	88.2	83.3	73.4	63.9	55.9
		MCWB	48.2	45.7	51.1	58.4	65.3	72.5	73.3	73.9	70.1	64.3	57.2	51.1
	5%	DB	46.5	47.4	54.0	65.8	74.8	81.9	86.3	84.3	79.2	69.2	60.1	51.2
		MCWB	42.0	40.8	45.5	54.7	62.6	69.7	72.5	71.5	68.0	62.4	53.3	46.6
	10%	DB	41.9	42.7	48.8	60.8	70.0	78.6	82.6	81.4	75.2	64.5	55.6	47.0
		MCWB	37.3	36.9	41.9	50.9	60.3	67.3	70.9	70.2	66.8	58.2	50.6	43.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.1	55.5	58.2	64.5	70.8	76.4	78.2	77.9	75.2	70.3	63.7	58.4
		MCDB	59.7	58.3	65.6	75.5	82.2	87.5	89.3	88.8	83.4	74.8	67.5	61.3
	2%	WB	48.9	47.1	52.5	60.4	67.8	74.1	76.1	75.7	72.7	66.0	59.2	52.6
		MCDB	51.9	51.4	59.9	70.3	77.8	83.6	85.8	84.0	79.4	71.1	62.4	55.5
	5%	WB	42.7	41.3	46.4	56.5	64.7	71.9	74.4	74.1	70.9	63.0	55.0	47.7
		MCDB	46.1	46.6	52.2	64.3	71.9	79.6	82.8	81.2	76.5	68.1	58.1	50.1
	10%	WB	37.7	37.8	42.6	51.5	61.6	69.6	72.7	72.5	68.9	59.9	51.5	43.7
		MCDB	41.2	41.7	47.8	58.4	68.7	76.1	80.2	78.7	74.0	64.0	55.2	47.0
Mean Daily Temperature Range	5% DB	MDBR	15.1	16.5	16.0	17.7	17.6	17.6	17.9	18.0	18.5	17.6	16.3	14.6
		MCDBR	20.0	21.5	23.1	26.7	25.7	22.0	21.6	20.9	22.1	21.9	20.4	18.6
		MCWBR	17.4	16.2	15.4	15.5	14.0	11.6	9.5	9.8	11.3	13.7	15.5	16.6
	5% WB	MCDBR	19.6	20.3	21.6	24.4	22.4	20.3	18.4	17.6	17.6	19.4	17.0	18.0
MCWBR		18.4	16.9	16.4	16.3	13.6	11.7	9.5	9.4	10.9	14.3	15.2	18.0	
Clear-Sky Solar Irradiance	taub	0.289	0.298	0.320	0.364	0.400	0.445	0.464	0.429	0.373	0.332	0.308	0.292	
	taud	2.514	2.506	2.504	2.412	2.350	2.257	2.209	2.295	2.434	2.553	2.578	2.574	
	Ebn at Noon	274	290	295	287	278	264	258	263	272	273	265	264	
	Edh at Noon	22	26	30	36	39	43	44	39	32	25	21	20	
All-Sky Solar Radiation	RadAvg	549	824	1146	1473	1642	1772	1872	1685	1350	892	601	451	
	RadStd	35	68	78	159	185	164	117	72	89	77	48	50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+1.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126
(47) Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+68

Nomenclature: See separate page